Rev E: USB-C 2S battery charging and regulated power distribution

Rev E. USB and power circuitry. Read CHECKS.md for validation status.

BQ25798 two-cell charger and power path

File: charger.kicad_sch

Charger bootstrap, limits and battery temperature

File: charger_support.kicad_sch

Always-on 3.3 V

File: always_on.kicad_sch

Power controls and fuel gauge

File: controls.kicad_sch

+5V_CORE regulated supply

File: buck_U5.kicad_sch

+5V_IO regulated supply

File: buck_U6.kicad_sch

+3V3_AUDIO regulated supply

File: buck_U7.kicad_sch

Hardware RGB current limit

File: rgb_power.kicad_sch

Power connectors, fuse and mechanical mounting

File: harness.kicad_sch

Sheet: /
File: power.kicad_sch

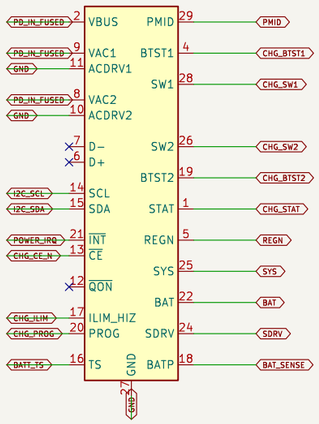
Title:

Size: A2 | Date: | **Rev:**

KiCad E.D.A. 9.0.7 | Id: 10/10

2S Li-ion only: 8.4 V termination, 6.04 kOhm PROG, 1.5 MHz.
Requires a protected, balanced 2S pack and 103AT-2 10k NTC on the cells. CE defaults disabled.
Firmware validates PD contract, programs limits, checks temperature and then enables charging.

U1
BQ25798RQMR



L1
1uH Isat>=12A

C1
10u 50V X7R

C2
10u 50V X7R

C3
10u 50V X7R

C4
10u 50V X7R

C5
10u 50V X7R

C6
10u 25V X7R

C7
10u 25V X7R

C8
10u 25V X7R

C9
10u 25V X7R

C10
10u 25V X7R

C11
10u 25V X7R

C12
10u 25V X7R

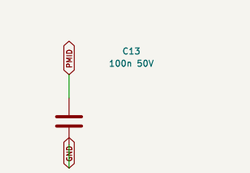
Connected schematic; hardware and manufacturing validation pending

Sheet: /BQ25798 two-cell charger and power path/
File: charger.kicad_sch

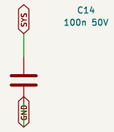
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Size: A3 | Date: | **Rev: E – engineering draft**
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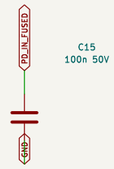
REGN divider limits default input current. A diode from INPUT_BOOST raises the hardware ceiling only after configuration.
TS uses 5.23k pullup and 30.1k parallel return with external 103AT-2. Open/short NTC inhibits charging.
No ship FET: SDRV has the specified 1nF capacitor.



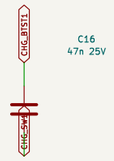
C13
100n 50V



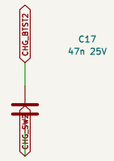
C14
100n 50V



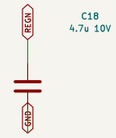
C15
100n 50V



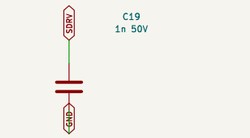
C16
47n 25V



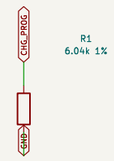
C17
47n 25V



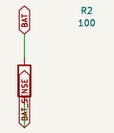
C18
4.7u 10V



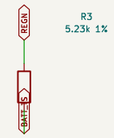
C19
1n 50V



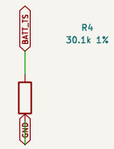
R1
6.04k 1%



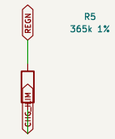
R2
100



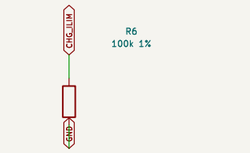
R3
5.23k 1%



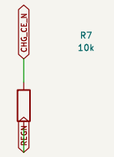
R4
30.1k 1%



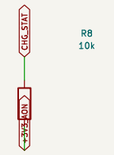
R5
365k 1%



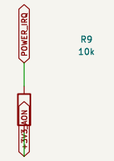
R6
100k 1%



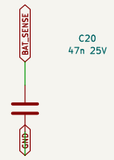
R7
10k



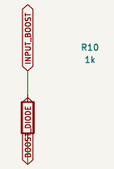
R8
10k



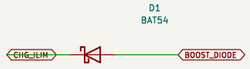
R9
10k



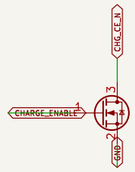
C20
47n 25V



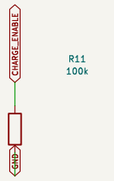
R10
1k



D1
BAT54



Q1
2N7002



R11
100k

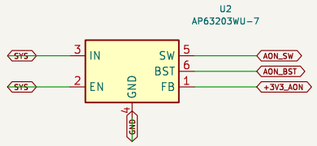
Connected schematic; hardware and manufacturing validation pending

Sheet: /Charger bootstrap, limits and battery temperature/
File: charger_support.kicad_sch

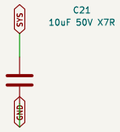
Title: Charger bootstrap, limits and battery temperature

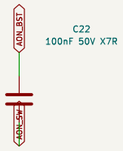
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KiCad E.D.A. 9.0.7		Id: 3/10

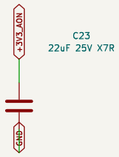
AP63203 fixed 3.3 V, 2 A regulator capacity; actual rail load must be measured.
VIN/EN are SYS rated. No low-voltage SLEEP pin is tied to the 2S rail.
Tight input/ground/bootstrap loops; short SW and Kelvin output feedback.

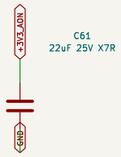












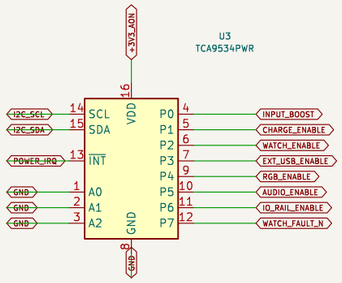
Connected schematic; hardware and manufacturing validation pending

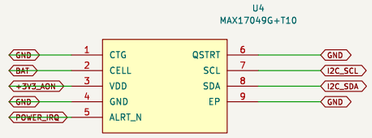
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File: always_on.kicad_sch

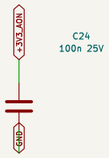
Title: Always-on 3.3 V: 2 A regulator

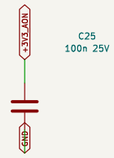
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KiCad E.D.A. 9.0.7		Id: 4/10

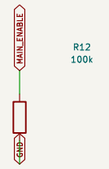
TCA9534 at 0x20; power-on pins are inputs with external pulldowns.
MAX17049 at 0x36 senses the protected 2S pack; its VDD is regulated 3.3 V.
MAIN_ENABLE is RP2040 GPIO23. POWER_REQUEST is the rear ON/OFF request.

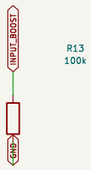


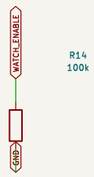


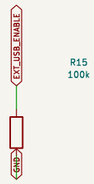


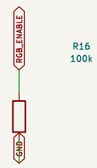


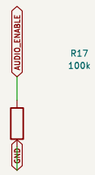


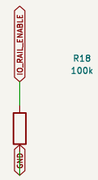












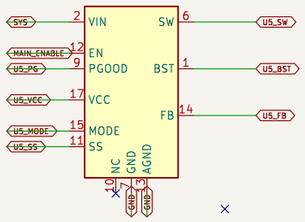
Connected schematic; hardware and manufacturing validation pending

Sheet: /Power controls and fuel gauge/
File: controls.kicad_sch

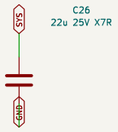
Title: Power controls and fuel gauge

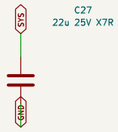
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KiCad E.D.A. 9.0.7		Id: 5/10

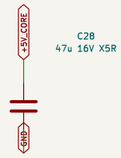
TPS568230, 600 kHz out-of-audio mode. Input is 2S system rail, never the 20 V PD input.
Power-current loops are routed locally; firmware enforces the shared input-power budget.

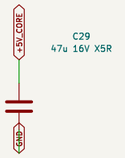


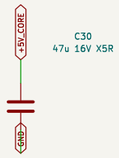


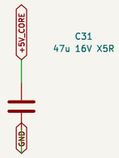


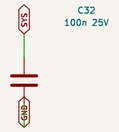


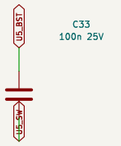


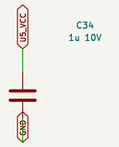


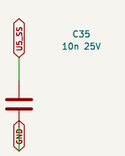


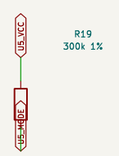


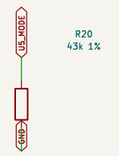


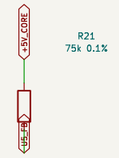


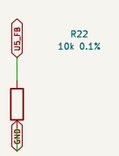


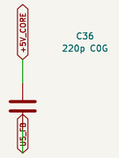


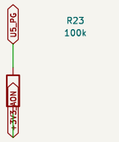












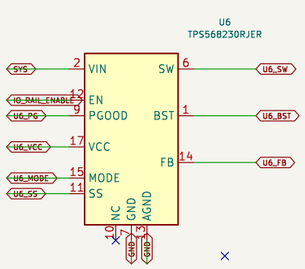
Connected schematic; hardware and manufacturing validation pending

Sheet: /+5V_CORE regulated supply/
File: buck_U5.kicad_sch

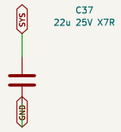
Title: +5V_CORE regulated supply

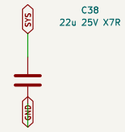
Size: A3	Date:	**Rev: E — engineering draft**
KiCad E.D.A. 9.0.7		Id: 6/10

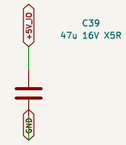
TPS568230, 600 kHz out-of-audio mode. Input is 2S system rail, never the 20 V PD input.
Power-current loops are routed locally; firmware enforces the shared input-power budget.

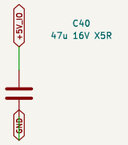


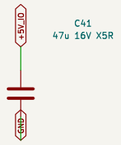




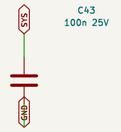


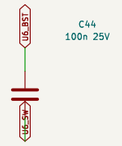


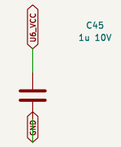


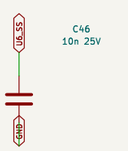


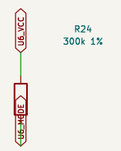


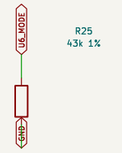


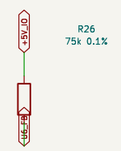


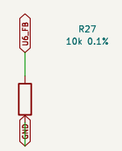


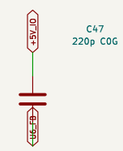


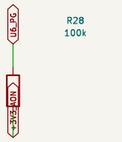












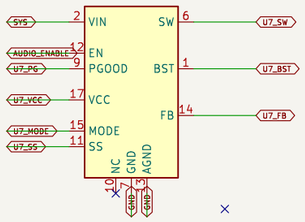
Connected schematic; hardware and manufacturing validation pending

Sheet: /+5V_IO regulated supply/
File: buck_U6.kicad_sch

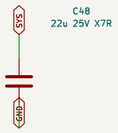
Title: +5V_IO regulated supply

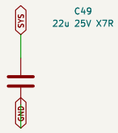
Size: A3 | Date: | **Rev: E – engineering draft**
KiCad E.D.A. 9.0.7 | Id: 7/10

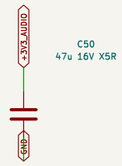
TPS568230, 600 kHz out-of-audio mode. Input is 2S system rail, never the 20 V PD input.
Power-current loops are routed locally; firmware enforces the shared input-power budget.

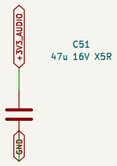


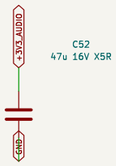


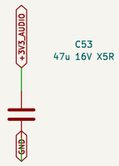


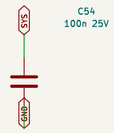


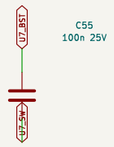


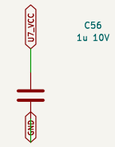


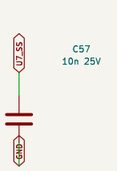


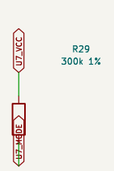


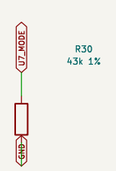


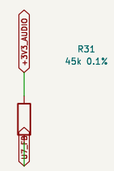


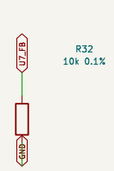


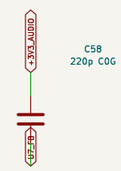


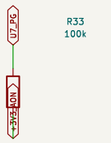












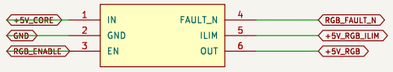
Connected schematic; hardware and manufacturing validation pending

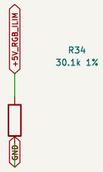
Sheet: /+3V3_AUDIO regulated supply/
File: buck_U7.kicad_sch

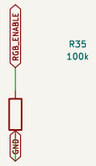
Title: +3V3_AUDIO regulated supply

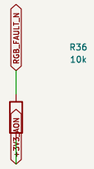
Size: A3 | Date: | **Rev: E – engineering draft**
KiCad E.D.A. 9.0.7 | Id: 8/10

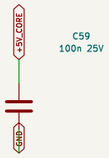
TPS2553 30.1k current limit protects the RGB rail and 1 A GH contact.
Software RGB budget is 0.7 A for 108 keys plus the battery pixel. Full-brightness all-white is not supported.

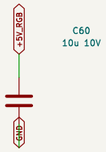












Connected schematic; hardware and manufacturing validation pending

Sheet: /Hardware RGB current limit/
File: rgb_power.kicad_sch

Title: Hardware RGB current limit

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Protected 2S pack: 8.4 V max, >=10 A discharge capability, factory balancing/protection.
J1 adapter power, J2 rear regulated power, J3 rear control, J4 keyboard power, J5 keyboard I2C, J6 CM4/display, J7 audio, J8 battery, J9 NTC.
Power connector ratings and harness wire size must cover each load. No connector shares raw PD with a 5 V rail.

J1
From rear J9

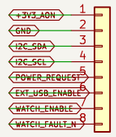


F1
3A 32V

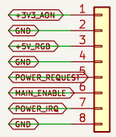
J2
To rear J10



J3
To rear J11



J4
To keyboard J3



J5
To keyboard J4 pins 3/4/5/6



J6
CM4 / display supply



J7
To audio J3 pins 1/2



J8
Protected 2S battery



F2
10A 32V

J9
Cell-mounted 103AT-2 NTC



H1
M2 clearance

H2
M2 clearance

H3
M2 clearance

H4
M2 clearance

#FLG126
PWR_FLAG

#FLG127
PWR_FLAG

#FLG128
PWR_FLAG

#FLG133
PWR_FLAG

#FLG130
PWR_FLAG

#FLG131
PWR_FLAG

#FLG132
PWR_FLAG

Connected schematic; hardware and manufacturing validation pending

Sheet: /Power connectors, fuse and mechanical mounting/
File: harness.kicad_sch

Title: Power connectors, fuse and mechanical mounting

Size: A3	Date:	Rev: E — engineering draft
KiCad E.D.A. 9.0.7		Id: 10/10